SECTION 4 – ADVANCED SERVICE

4–1 THEORY OF OPERATION

4–1–1 General

Lightning PDU is the primary interface to facility power. It converts input line voltage to 208 VAC (3 phase) or 120 VAC (1 phase) as required by each MR subsystem. It provides protection against high voltage spikes caused by lighting or switching transients and both transverse mode and common mode high frequency noise.

Regulated models provide active powerline regulation and protection against powerline surges and sags.

Lightning PDU is a three–phase tapped voltage regulator. It consists of a multi–tapped transformer. Depending on input voltage, output voltage is regulated by selecting an appropriate transformer tap.

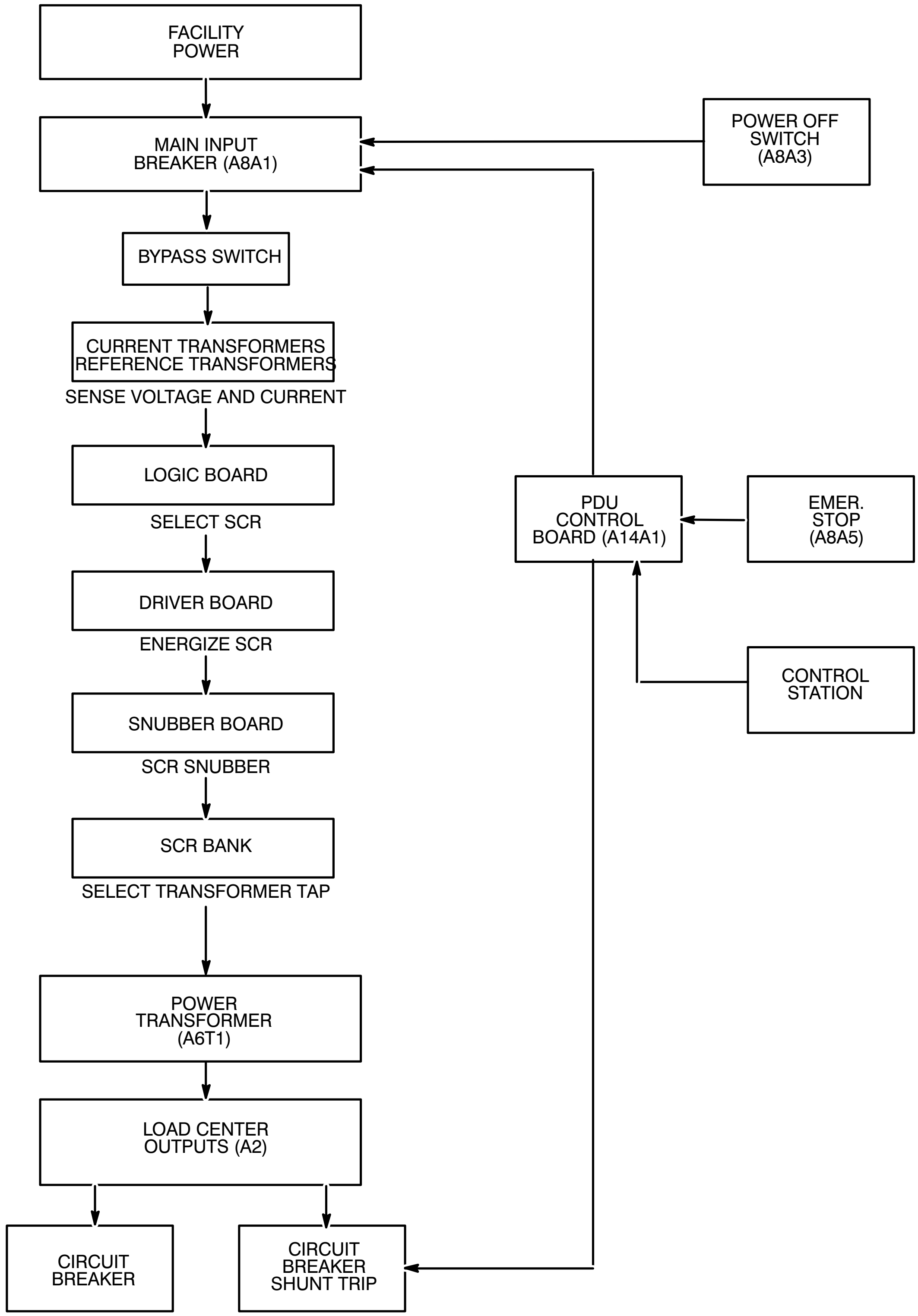
In models 46–317099 P20 and P21, tap selection is done electronically. There are 6 transformer taps per phase. Taps are spaced at 3% intervals of nominal input voltage. Tap 5 is located at the nominal input voltage to within 1.5% of nominal typically.

Illustration 4–1 shows the voltage regulation process. Each input line–to–line voltage is monitored and compared to a reference voltage. Each line current is monitored and compared to a preset level in order to activate the regulation circuit.

The logic board (A14A2, A14A3, A14A4) determines if a voltage correction is required. It selects the appropriate tap, and signals the driver board when to turn off the conducting SCR and when to turn on a new SCR.

The 6 SCRs per phase are mounted on a heat sink. They are fused together by 18–gauge copper bus wire (fuselinks). At any time only one SCR is conducting. If for any reason two or more SCRs begin to conduct, then one or more fuselinks will open. The SCR sense board monitors fuselink continuity and shunt trips the main circuit breaker if this circuit is open.

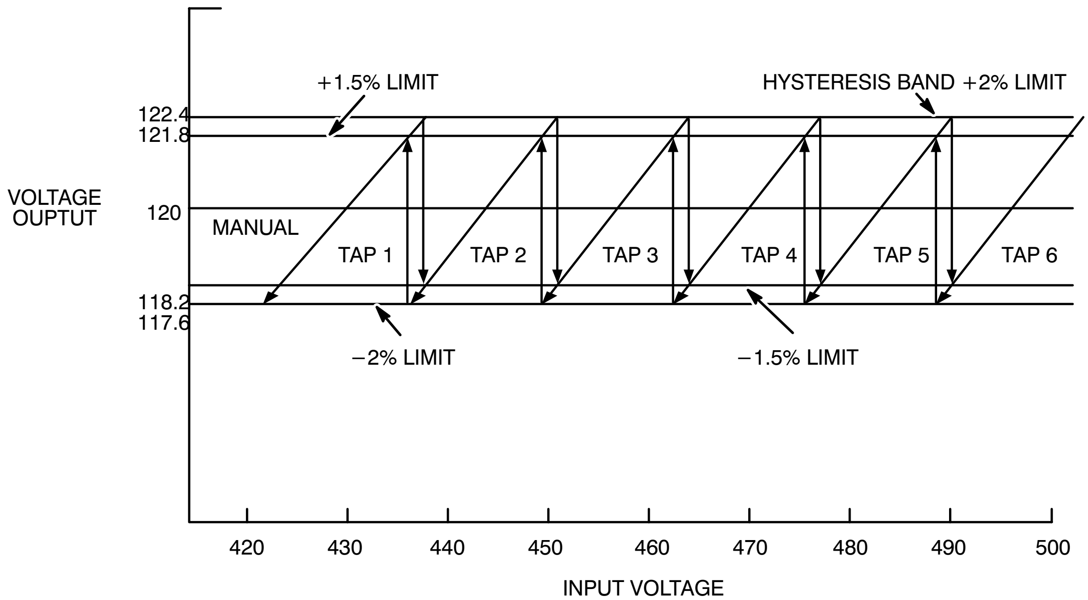
4–1–1 General (continued)



LIGHTNING PDU BLOCK DIAGRAM
ILLUSTRATION 4–1

4–1–1 General (continued)

Illustration 4–2 is a model of regulator output accuracy as a function of input voltage. Output voltage is corrected to within 1.5% of nominal input voltage. However, when switching to an adjacent tap, a hysteresis band is designed into the logic circuitry to prevent flip–flopping between taps. Therefore output voltage is always within 2% of nominal voltage.



REGULATOR OUTPUT ACCURACY
ILLUSTRATION 4–2

Section 5 is the schematic for regulation–type Lightning PDUs. An input filter assembly:

- contains an inductor which removes peaks of voltage spikes,
- reduces rate of rise of voltage (a rapid voltage rate of rise damages SCRs), and
- removes high frequency noise from line power.

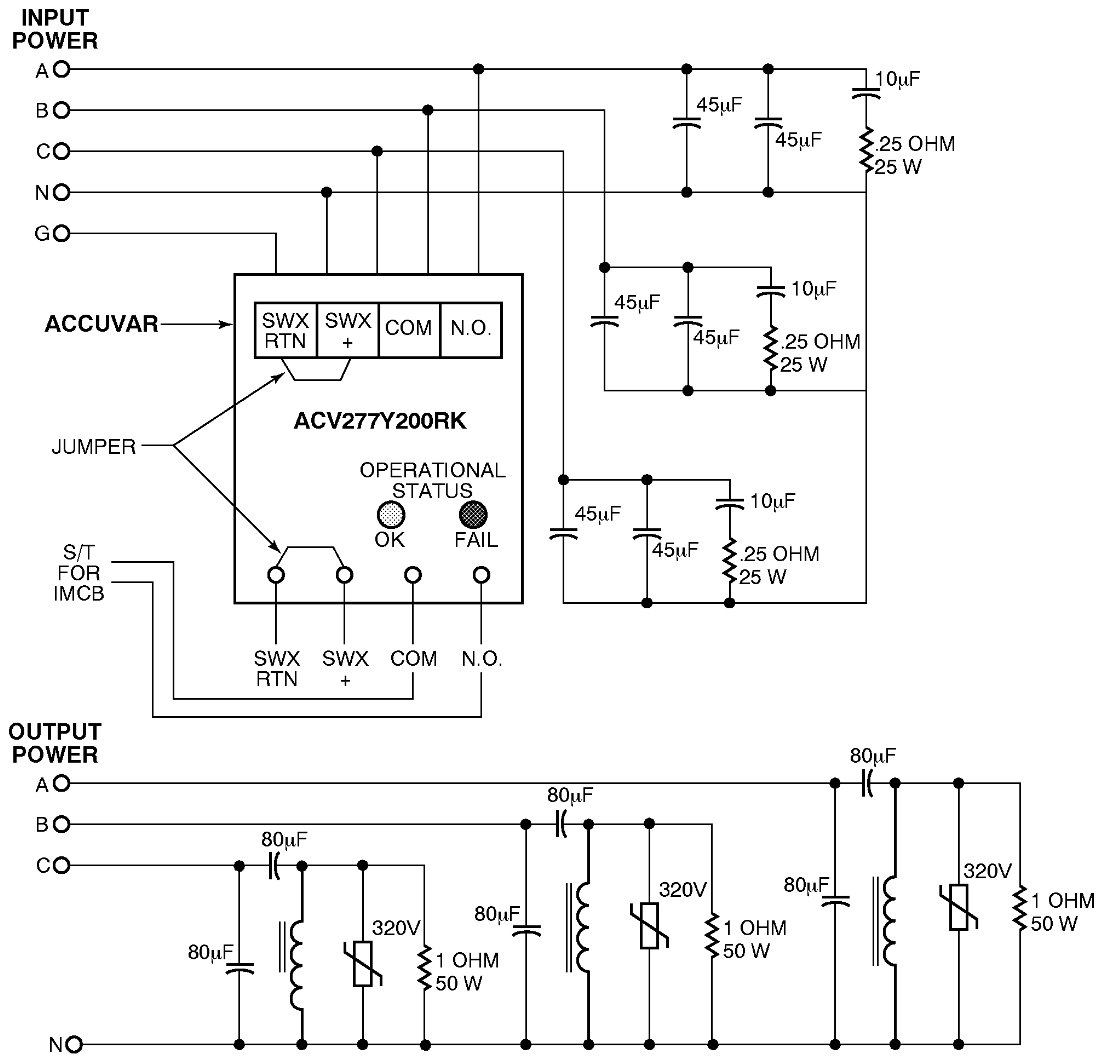
All Lightning PDU models have capacitors (A10) across output line–to–neutral terminals. The capacitors smooth out any waveform distortion that occurs because of tap switching.

4–1–2 Filter Assembly

The filter assembly is a brute force low pass filter. The filter assembly provides for high–energy surge suppression. Clamped high–energy transients are attenuated to acceptable levels. The filter contains a tuned damping circuit to reduce overshoot. Filter output impedance is designed to minimize voltage waveform distortion due to pulsed loading.

4−1−2 Filter Assembly (continued)

The three basic filter components are the line filter, the shunt module and the output filter. These components provide attenuation of line noise and high−energy surges. See Illustration 4−3.



FILTER ASSEMBLY BLOCK DIAGRAM
ILLUSTRATION 4−3

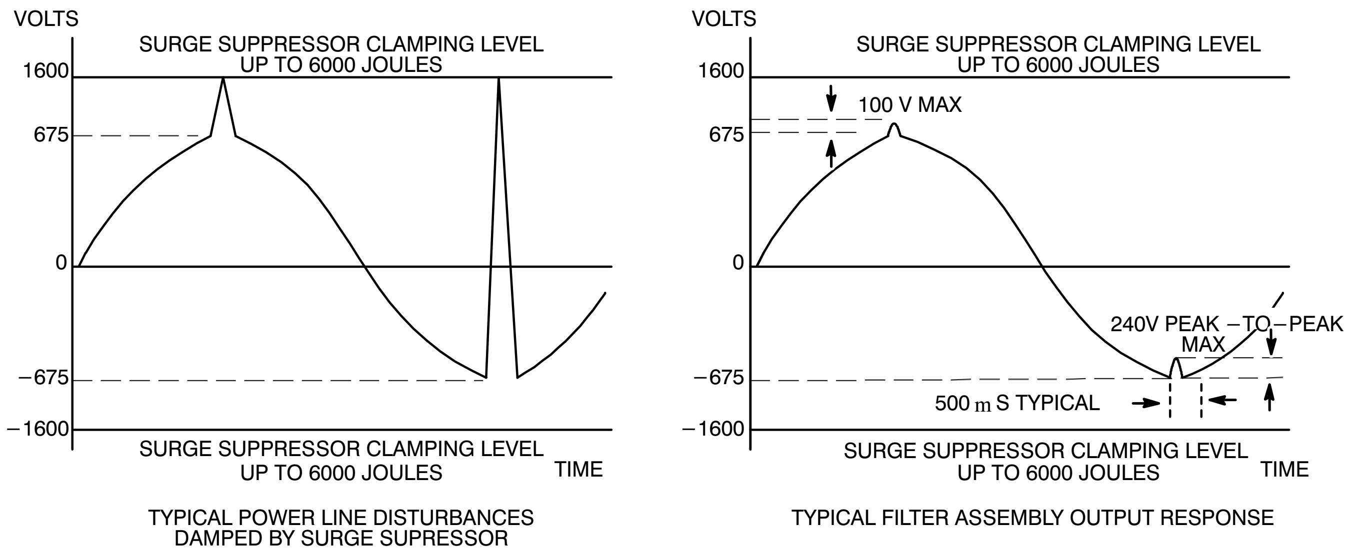
Surge Suppressor. The surge suppressor is located at the input to the filter. An Accuvar is used for surge suppression. The surge suppressor limits peak voltage passed on the filter, and subsequently, to the load. When a surge exceeds the suppressor clamping level, the suppressor conducts and diverts surge currents away from the load. Current flow into the surge suppressor limits the surge voltage. Power is dissipated by the suppressor for the duration of the surge.

Line Filters The line filter removes all high frequency noise with capacitors and series capacitors/resistors networks.

Load Filters The load filter removes high frequency noise with capacitors, inductor/capacitors, and resistant/capacitors circuits.

4–1–2 Filter Assembly (continued)

Typical power line disturbance and filter response are shown in Illustration 4–4.



TRANSIENT RESPONSE
ILLUSTRATION 4–4

4–1–3 Logic Board

Section 5–3 shows the electrical schematic for the logic board. Logic board functions are the following:

Load Level Detector. The load level detector senses line current using a current transformer and signal applied at connector P1, and pins 12 and 13 of U19. Lightning PDU is released to regulate when voltage across C42 is greater than 8 VDC. U19 amplifies current signal using output on pin 14. The current signal is rectified by CR30. Voltage across C42 is adjusted by potentiometer R62. Adjust R62 so that voltage across C42 is greater than 8 VDC when on tap 6 of SCR.

Voltage Detector. The voltage detector receives input voltage signal from reference transformers T2, T3, T4. This signal is compared with a reference level determined by R39. The output of op amp U19 pin 7 is fed to the comparator network. The voltage detector generates a clock signal when the input voltage is at its negative peak (270). U6 outputs a square wave in phase with line voltage at pin 7. This turns on timer (U3). Timer pin 3 stays high for 11.5 ms.

Comparator Network. The comparator network consists of op amps U6 and U11, and resistors R15, R28, R29, R35, R36, R42, and R41. This network correlates with taps on transformer winding. The signal on U19 pin 7 is compared to select appropriate transformer taps. Output of the comparator network is fed to voltage flip–flop U10.

Voltage Flip–Flop. The voltage flip–flop, U10, transfers input tap selection information to output when it receives a clock signal on pin 9. This clock signal comes from change detector circuit U13 pin 3. This clock signal is generated at the 270–degree phase angle of the voltage sine wave by U3 at pin 3.

4−1−3 Logic Board (continued)

Change Detector Circuit. The change detector circuit monitors input and output of the U10 flip−flop once per cycle. When U10 input is different from output, the circuit generates three outputs:

1. The output on U5 pin 4 turns off all SCR drive circuits. The conducting SCR continues to conduct until current becomes zero.

2. The output on U13 pin 3 transfers tap selection information from input to output on voltage flip−flop U10.

3. The output on U13 pin 11 or pin 8 provides a clock signal to output flip−flop U8.

Output Stage. Output flip−flop U8 holds information passed by voltage flip−flop U10 until it receives a clock signal from U15 pin 8 or 11. The output stage is the interface between logic board and driver board.

Master Reset. The master reset circuit is activated when the main circuit breaker is turned on. Output is inhibited for 6 seconds. This circuit is also triggered by the undervoltage detection circuit.

Undervoltage Detection Circuit. This circuit compares the input voltage signal to a preset reference of 3 V. When voltage falls below the reference level U19 pin 1 becomes low. This drives pins 9 and 12 of U2 low, which turns off the SCR drive circuit. Undervoltage shutdown occurs when line voltage sags by 40% for at least 2 cycles.

4−1−4 Driver Board

Section 5−4 is the driver board schematic. The driver board has a 5 KHz oscillator (U3). It provides a push/pull signal to MOSFETS which energizes the pulse transformer primaries. Pulse transformer output is a square wave. This signal is applied to the SCR bank through snubber boards.

4−1−5 PDU Control Board

Section 5−6 is the PDU control board schematic. The PDU control board allows exterior controls to shunt trip load panel circuit breakers (A5). These connection points are J1 (coil), J2 (system cabinet status) and J3 (system cabinet PDU remote). The PDU control board also interfaces the emer. stop reset function. Activation of emergency stop from magnet enclosure, console, or the remote control panel located on system cabinet, trips gradient amplifier cabinets, RF amplifier cabinet, and Gram cabinet load panel circuit breakers. It also trips any option circuit breakers which have shunt trips, such as the resistive shim power supply. The emergency stop reset switch is located on the front panel (A8A5).

Note

Whenever fuse F1 or F8 opens, the Gradient Amplifier Cabinets, Gram Cabinet, RF Amplifier Cabinet and optional Load Panel Circuit Breakers with Shunt Trips are tripped.

4−1−6 Snubber Board

Section 5−7 is the snubber board schematic.

4−1−7 SCR Sense Board

Section 5−5 is the SCR sense board schematic.